

EGCI341 Final Exam Review

About Final Exam

Final Exam

- Date: Th. 16th July, 2026
- Time: 12:00-13:50
- Room: 1314, MUIC Building
- Closed-book Exam
- 1-page A4, single side cheat-sheet is allowed
- Don't forget calculator (Not allow to borrow calculator)
- Student ID and student uniform
- Do not use red color pen
- Pencil with minimum 2B

Final For Term : 3/2025	DISABLE EXAM
Date Time	16/07/2026 Time : 12:00 - 13:50
Status	Approved
Exam Room	1314 Lecture Room
Proctor #	2
Requests	☒ Lecturer Present ☒ Exam Calculator
Student Remarks	- 1-page A4 single side cheat sheet is allowed. - Write student id on your cheat sheet.

Topics

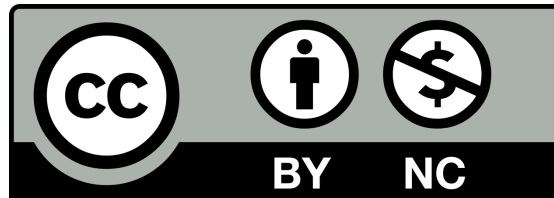
Opensource Software License– Software Cost Estimation (8 Questions)

- Opensource Software License (1 short answer)
- Software Testing (4 main questions, 1 short answer)
- Software Integration (1 short answer)
- Project Management (1 main question, 2 short answers)
- Software Cost Estimation (2 main questions. 1 short answer)

Opensource Software License

Software Reuse: Short Answers

- If the GNU opensource software is integrated with the software that currently the team is developing. When this software is done, Can the team use Apache license for their own software.
- What is the meaning of the figure below?



- Do the source code's owners of opensource software license take responsibility on their code if another programmer integrates these code to a new project?

DevOps



DevOps

- DevOps Process & Tools
- Containerization
- DevOps Solution for one scenario

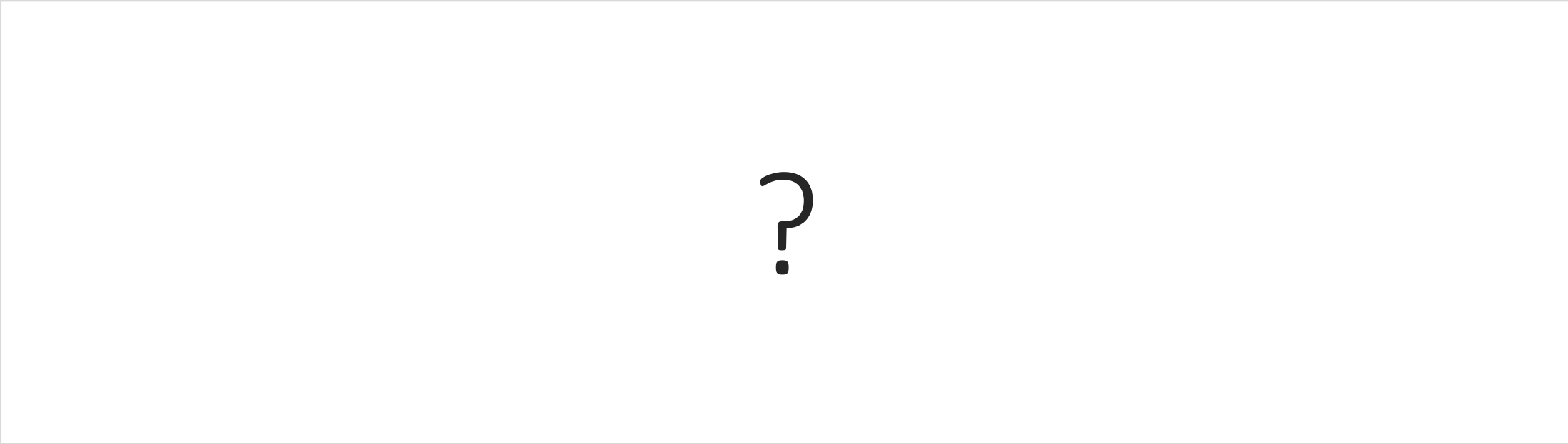
Project Management

Project management

	Task	Duration	Start	End	Pred
A	Identify Interviewee	5	3-Sep-2018	7-Sep-2018	
B	Conduct Interviews	30	10-Sep-2018	19-Oct-2018	A
C	Review Documentation	10	22-Oct-2018	02-Nov-2018	A,B
D	Analyze Data	10	5-Nov-2018	16-Nov-2018	C
E	Draft Preliminary Report	20	19-Nov-2018	14-Dec-2018	D
F	Obtain Feedback on Draft Report	10	17-Dec-2018	28-Dec-2018	E
G	Write Final Report	5	31-Dec-2018	4-Jan-2019	F
H	Prepare Presentation	10	31-Dec-2018	11-Jan-2019	D, F
I	Make Final Presentation	1	14-Jan-2019	14-Jan-2019	H, G

Project management (cont.)

Draw PERT Diagram



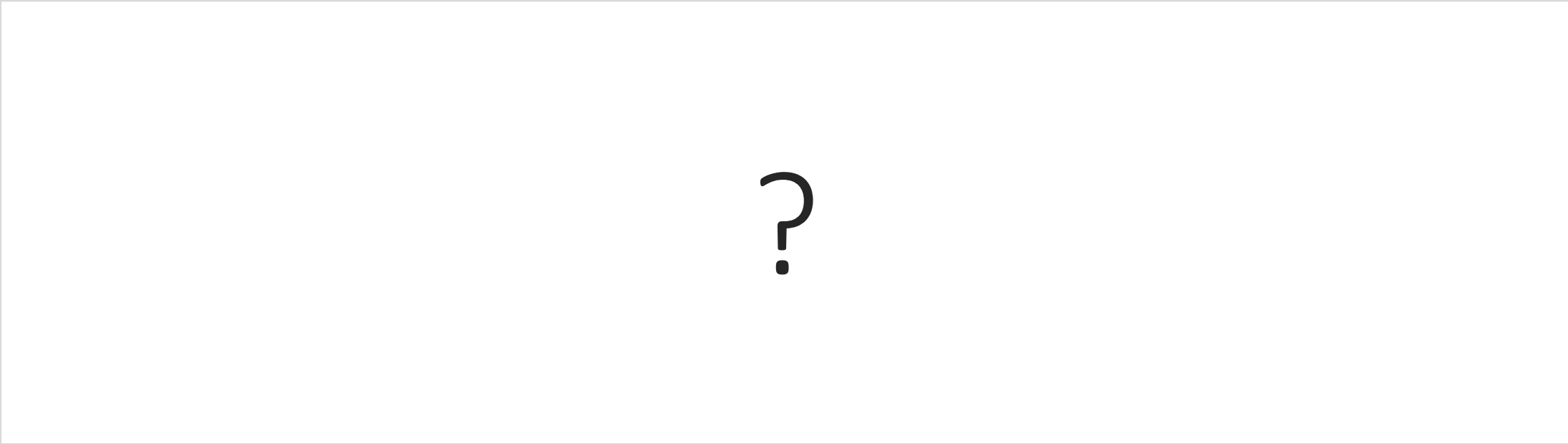
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Project management (cont.)

- Find critical path
- Find slack time

Project management (cont.)

Draw GANNT Chart

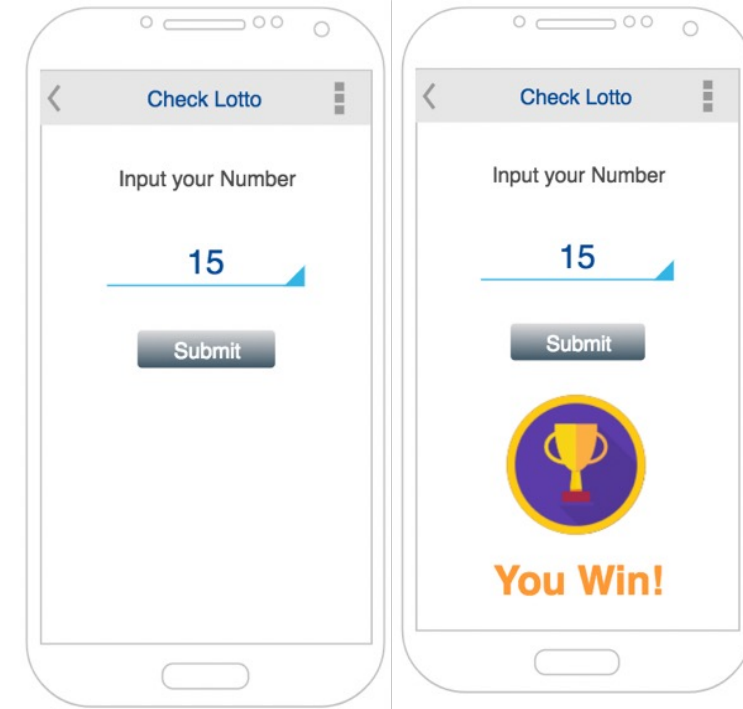


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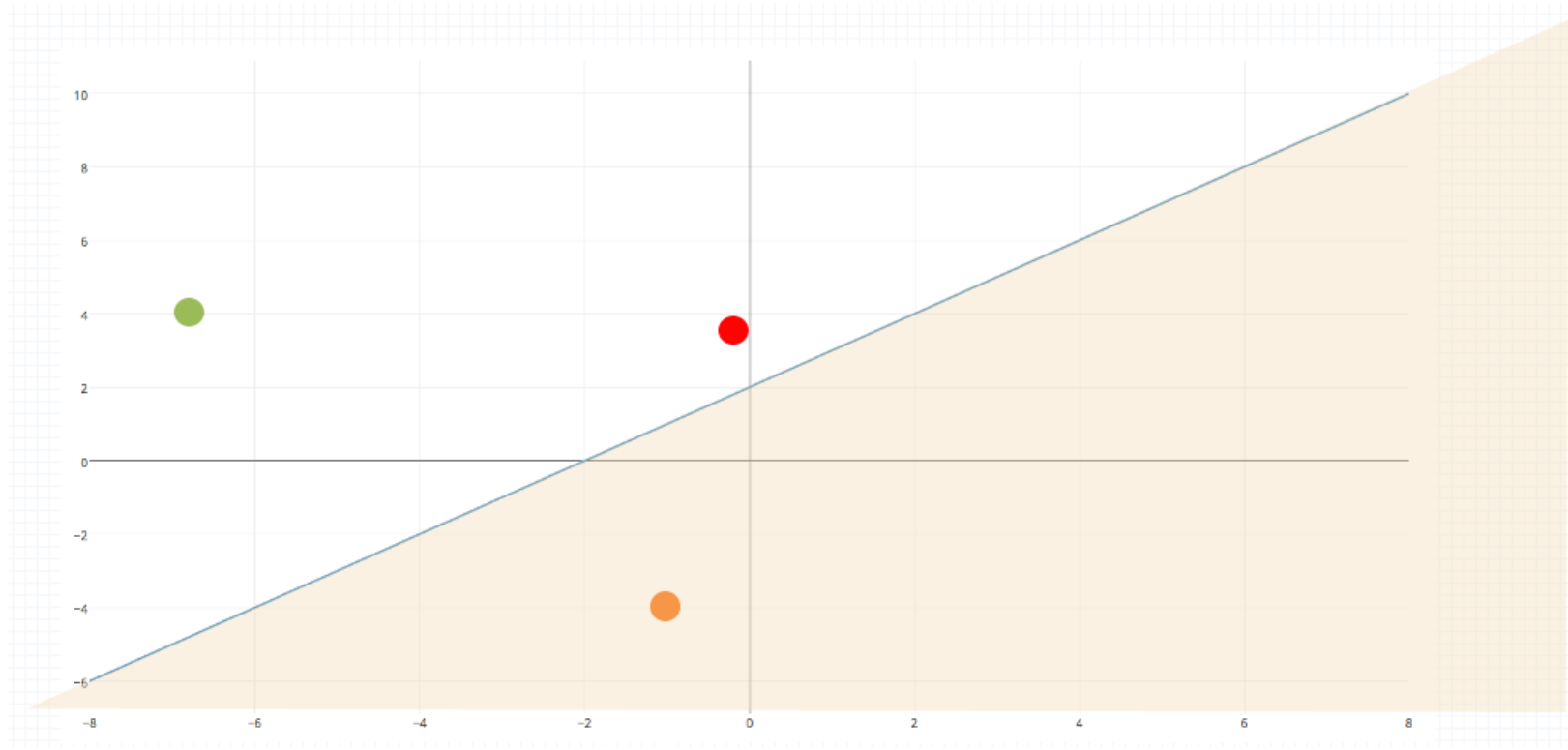
Software Testing

Software Testing: Check lotto no. application

- Design the test case including the testing input for a black-box and white-box testing.
- In this application, user can input only two-digit no. from 00 to 99 into the application to check whether the input no. matches with one of winning numbers. For example, if the winning no. are 03, 18, 46, 69, 84, then the input no. must be matched to one of these winning no. to win the lotto.



Design the Black-box and White-box Testing for the Graph Below



The application can classify whether the point (x,y) is inside the yellow area. If $y \leq x+2$, the point (x,y) is inside the yellow area. The input of this application is the point or coordinate (x,y) . x and y are integers. The output of this application is 'true' when the point (x,y) is inside the yellow area and 'false' when the point is outside the yellow area.

Software Cost Estimation

COCOMO II

6.1 COCOMO II Effort Equation

As an example, a project with all Nominal Cost Drivers and Scale Drivers would have an M of 1.00 and exponent is equal 1.0997. Assuming that the project is projected to consist of 8,000 source lines of code. $A = 2.94$

Find COCOMO II estimation?

6.2 Effort Adjustment Factor (referring from 1)

The Effort Adjustment Factor in the effort equation is simply the product of the effort multipliers corresponding to each of the cost drivers for your project.

For example, if your project is rated Very High for Complexity, and Low for Language Experience, and all of the other cost drivers are rated to be Nominal and $A = 2.94$

Find the multiplier (M) and find COCOMO II estimation?

Cost Drivers	Ratings					
	Very Low	Low	Nominal	High	Very High	Extra High
Product attributes						
Required software reliability	0.75	0.88	1.00	1.15	1.40	
Size of application database		0.94	1.00	1.08	1.16	
Complexity of the product	0.70	0.85	1.00	1.15	1.30	1.65
Hardware attributes						
Run-time performance constraints			1.00	1.11	1.30	1.66
Memory constraints			1.00	1.06	1.21	1.56
Volatility of the virtual machine environment		0.87	1.00	1.15	1.30	
Required turnabout time		0.87	1.00	1.07	1.15	
Personnel attributes						
Analyst capability	1.46	1.19	1.00	0.86	0.71	
Applications experience	1.29	1.13	1.00	0.91	0.82	
Software engineer capability	1.42	1.17	1.00	0.86	0.70	
Virtual machine experience	1.21	1.10	1.00	0.90		
Programming language experience	1.14	1.07	1.00	0.95		
Project attributes						
Application of software engineering methods	1.24	1.10	1.00	0.91	0.82	
Use of software tools	1.24	1.10	1.00	0.91	0.83	
Required development schedule	1.23	1.08	1.00	1.04	1.10	

Practice

No. of Functions	Description	Weight
2	Read text files	Average
3	Write data to csv files	Low
3	Query data from database	High
1	Get name and address from user	Low
3	Print reports on the screen	Average
2	Print reports to the printer	Average
2	Read data from sensors	Low

1. Find function points
2. Find estimated LOC
3. Find cost of software
4. Find Effort

- TDI: 48
- Language: Visual C++
- LOC/P-month: 360
- \$/LOC: 24